

FRAMEWORKS OF COLLECTION(30/7/26)

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1.LOGGER

```
1 interface Logger {
2     void debug(String message);
3     void info(String message);
4     void warning(String message);
5     void error(String message);
6 }
7
8 public class Log4jLogger implements Logger {
9
10    public void debug(String message) {
11        System.out.println("DEBUG: " + message);
12    }
13
14    public void info(String message) {
15        System.out.println("INFO: " + message);
16    }
17
18    public void warning(String message) {
19        System.out.println("WARNING: " + message);
20    }
```

```
22    public void error(String message) {
23        System.out.println("ERROR: " + message);
24    }
25
26    public static void main(String[] args) {
27        Log4jLogger logger = new Log4jLogger();
28
29        logger.debug("Application started");
30        logger.info("User logged in");
31        logger.warning("Low memory");
32        logger.error("File not found");
33    }
34 }
```

OUTPUT

```
DEBUG: Application started
INFO: User logged in
WARNING: Low memory
```

2.CHARACTER

```
1 interface Character {
2     void move();
3     void attack();
4     void takeDamage(int damage);
5 }
6
7 public class Hero implements Character {
8
9     public void move() {
10         System.out.println("Hero is moving.");
11     }
12
13     public void attack() {
14         System.out.println("Hero attacks with a sword.");
15     }
16
17     public void takeDamage(int damage) {
18         System.out.println("Hero takes " + damage + " damage.");
19     }
20
```

```
21     public static void main(String[] args) {
22         Hero hero = new Hero();
23         Villain villain = new Villain();
24         Monster monster = new Monster();
25
26         hero.move();
27         hero.attack();
28         hero.takeDamage(20);
29
30         villain.move();
31         villain.attack();
32         villain.takeDamage(30);
33
34         monster.move();
35         monster.attack();
36         monster.takeDamage(15);
37     }
38 }
39
40 class Villain implements Character {
```

```

42     public void move() {
43         System.out.println("Villain is moving.");
44     }
45
46     public void attack() {
47         System.out.println("Villain attacks with dark magic.");
48     }
49
50     public void takeDamage(int damage) {
51         System.out.println("Villain takes " + damage + " damage.");
52     }
53 }
54
55 class Monster implements Character {
56
57     public void move() {
58         System.out.println("Monster is moving.");
59     }
60
61     public void attack() {
62         System.out.println("Monster attacks with claws.");
63     }
64
65     public void takeDamage(int damage) {
66         System.out.println("Monster takes " + damage + " damage.");
67     }
68 }

```

OUTPUT

```

Hero is moving.
Hero attacks with a sword.
Hero takes 20 damage.
Villain is moving.
Villain attacks with dark magic.
Villain takes 30 damage.
Monster is moving.
Monster attacks with claws.
Monster takes 15 damage.

```

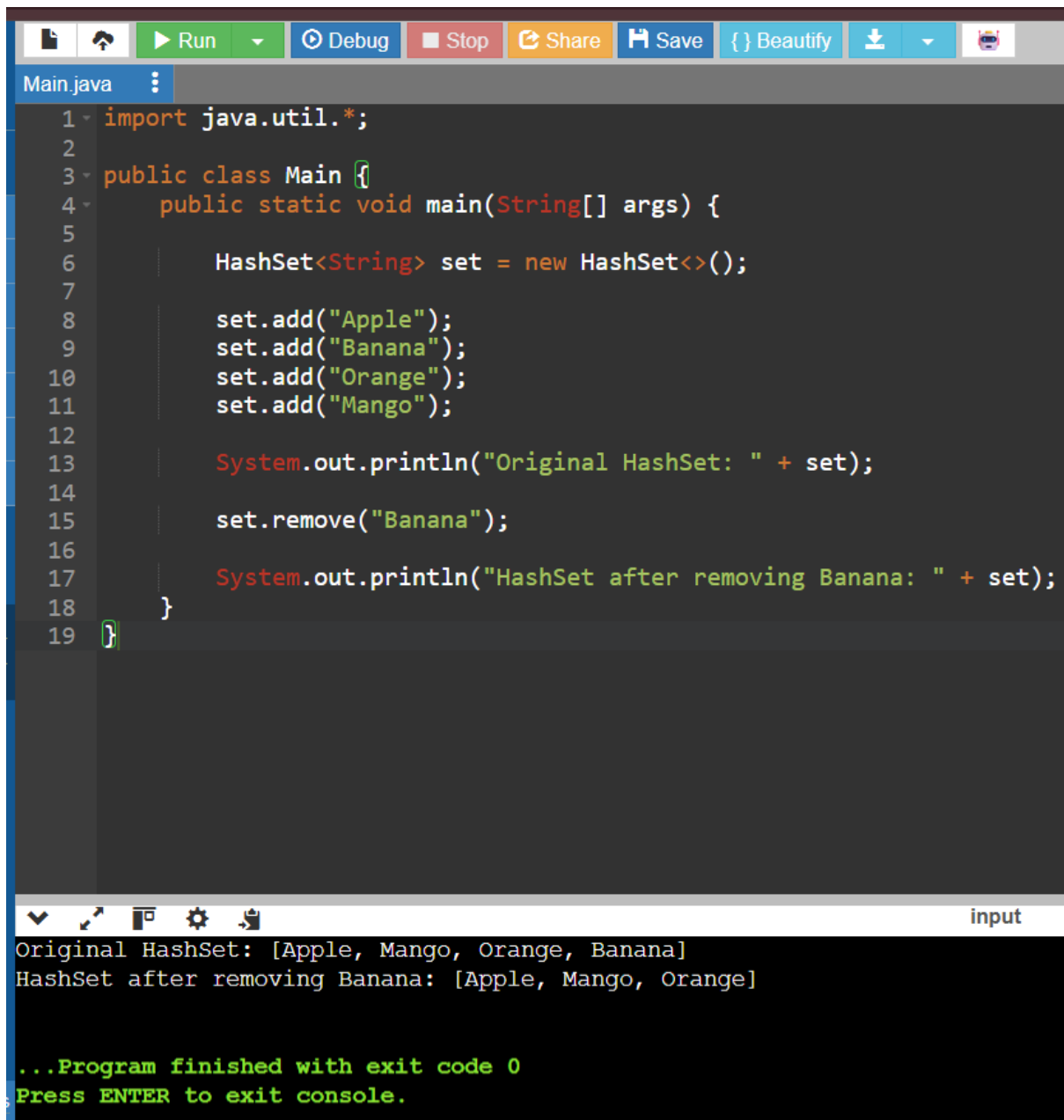
3.SORT

```
1 import java.util.*;
2
3 public class Main {
4     public static void main(String[] args) {
5
6         List<Integer> list = new ArrayList<>();
7
8         list.add(45);
9         list.add(12);
10        list.add(78);
11        list.add(23);
12        list.add(5);
13
14        System.out.println("Original List: " + list);
15
16        Collections.sort(list);
17
18        System.out.println("Sorted List: " + list);
19    }
20 }
```

OUTPUT

```
Original List: [45, 12, 78, 23, 5]
Sorted List: [5, 12, 23, 45, 78]
```

4.HASHSET



```
1 import java.util.*;
2
3 public class Main {
4     public static void main(String[] args) {
5
6         HashSet<String> set = new HashSet<>();
7
8         set.add("Apple");
9         set.add("Banana");
10        set.add("Orange");
11        set.add("Mango");
12
13        System.out.println("Original HashSet: " + set);
14
15        set.remove("Banana");
16
17        System.out.println("HashSet after removing Banana: " + set);
18    }
19 }
```

Original HashSet: [Apple, Mango, Orange, Banana]
HashSet after removing Banana: [Apple, Mango, Orange]

...Program finished with exit code 0
Press ENTER to exit console.

5.TREEMAP

```
1 import java.util.*;
2
3 public class Main {
4     public static void main(String[] args) {
5
6         TreeMap<String, Integer> map = new TreeMap<>();
7
8         map.put("Apple", 10);
9         map.put("Banana", 20);
10        map.put("Orange", 15);
11        map.put("Mango", 25);
12
13        System.out.println("TreeMap Entries:");
14
15        for (Map.Entry<String, Integer> entry : map.entrySet()) {
16            System.out.println(entry.getKey() + " = " + entry.getValue());
17        }
18    }
19 }
```

OUTPUT

```
TreeMap Entries
Apple = 10
Banana = 20
Mango = 25
Orange = 15
```

6.PRIORITY QUEUE

```
1 import java.util.*;
2
3 public class PriorityQueueExample {
4     public static void main(String[] args) {
5         PriorityQueue<Integer> pq = new PriorityQueue<>();
6
7         pq.add(30);
8         pq.add(10);
9         pq.add(20);
10        pq.add(40);
11
12        System.out.println(pq);
13    }
14 }
```

OUTPUT

```
[10, 30, 20, 40]
```

7.LINKEDHASHMAP

```
1 import java.util.*;
2
3 public class Main {
4     public static void main(String[] args) {
5
6         LinkedHashMap<String, Integer> map = new LinkedHashMap<>();
7
8         map.put("Apple", 10);
9         map.put("Banana", 20);
10        map.put("Orange", 15);
11        map.put("Mango", 25);
12
13        System.out.println("Keys in insertion order:");
14
15        for (String key : map.keySet()) {
16            System.out.println(key);
17        }
18    }
19 }
```

OUTPUT

```
Keys in insertion order:
Apple
Banana
Orange
Mango
```

8.MAX MIN

```
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Set<Integer> set = new HashSet<>();
5
6         set.add(25);
7         set.add(10);
8         set.add(45);
9         set.add(5);
10        set.add(30);
11
12        System.out.println("Set: " + set);
13
14        int max = Collections.max(set);
15        int min = Collections.min(set);
16
17        System.out.println("Maximum value: " + max);
18        System.out.println("Minimum value: " + min);
19    }
20 }
```

OUTPUT

```
Set: [5, 25, 10, 45, 30]
Maximum value: 45
Minimum value: 5
```